

Aislinn Coleman-Plante

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EDUCATION

University of Colorado Boulder

Bachelor of Arts, Major: Astronomy | Minor: French
Honors: Magna cum laude

May 2024
GPA: 3.75

EXPERIENCE

Post-baccalaureate Research Fellow, National Radio Astronomy Observatory

Sept 2024-Present

Supervisor: John Tobin, PhD

Research Assistant, CU Boulder

May 2023-May 2024

Supervisor: Meredith MacGregor, PhD
Honors thesis presented Spring 2024, to be submitted for publication

Learning Assistant, Modern Physics: CU Boulder

Spring 2022

Topics: Special Relativity, Quantum Mechanics, Atomic Structure
Supervisor: Noah Finkelstein, PhD

POSTERS & PUBLICATIONS

Coleman-Plante A, Tobin J, “**Multiplicity in the Early Phases of Star Formation: The Taurus Molecular Cloud,**” iPoster, presented at 245th Meeting of the American Astronomical Society, National Harbor, MD, January 2025.
<https://ui.adsabs.harvard.edu/abs/2025AAS...24524916C>

Coleman-Plante A, MacGregor M, “**Investigating the HD 139664 Debris Disk and Star with ALMA,**” iPoster, presented at 243rd Meeting of the American Astronomical Society, New Orleans LA, January 2024.
<https://ui.adsabs.harvard.edu/abs/2024AAS...24341103C>

Coleman-Plante A, “**Investigating the HD 139664 Debris Disk and Star with ALMA,**” Undergraduate honors thesis, 2024, University of Colorado Boulder.
https://scholar.colorado.edu/concern/undergraduate_honors_theses/5h73px85p

“**Coronal Heating as Determined by the Solar Flare Frequency Distribution Obtained by Aggregating Case Studies,**” James Mason [..., Aislinn Coleman-Plante] et al., Astrophysical Journal 2023.
<https://doi.org/10.48550/arXiv.2305.05687>

* Contributed through course-based research project; developed Python tools for baseline subtraction and flare energy analysis

RESEARCH PROJECTS

Using JWST to study Accretion in Orion Protostars

Sept 2025-Present

Supervisor: John Tobin, PhD

- Extract one-dimensional spectra of individual protostars from NIRSpec and MIRI data from the JWST HEFE large program
- Analyze atomic hydrogen lines to test for magnetospheric accretion and use line luminosities to obtain a more direct measurement of the disk-to-protostar accretion rates

Multiplicity in the Early Phases of Star Formation: The Taurus Molecular Cloud

Sept 2024-Present

Supervisor: John Tobin, PhD

- Developed pipelines for calibrating and imaging high-volume datasets from ALMA and VLA
- Conducted in-depth analysis of protostar population statistics
- Utilized ALMA molecular line data to trace Keplerian gas kinematics to constrain protostar masses
- Served as PI on a Cycle 12 ALMA proposal to expand the survey to Flat Spectrum sources; proposal was not accepted but ranked in the top quartile of distributed peer review

Investigating the HD 139664 Debris Disk and Star with ALMA

Honors thesis, Supervisor: Meredith MacGregor, PhD

Spring 2024

- Calibrated and reduced ALMA data using CASA
- Leveraged MCMC functions to model the assymetric architecture of this debris disk
- Iteratively modeled stellar luminosity, leading to the discovery of the first millimeter flares from an F-type star

Evolution of Stellar Activity as a Function of Age

Presented in Observations & Instrumentation II, Professor: Kevin France, PhD

Spring 2024

- Investigated five stars of varying ages with the Hubble Advanced Spectral Products program to analyze emission fluxes and track FUV evolution with stellar age
- Analyzed Hubble COS/STIS data and modeled emission lines to calculate luminosities
- Applied findings to an atmospheric mass-loss model for a 17 Myr hot Jupiter exoplanet

Multi-wavelength Observations of Flares on V1005

Presented in Observations & Instrumentation, Professor: Zachary Berta-Thompson, PhD

Fall 2023

- Collaborated in operating three telescopes at the Sommers-Bausch Observatory to identify stellar flares across different wavelengths and estimate temperatures using the Planck function
- Created pipeline to perform differential photometry and generate light curves in real-time

Modeling of Exoplanetary Transits in Multiple Wavelengths, using Titan Occultation Data

Presented in Research Methods in Astronomy, Professor: Paul Hayne, PhD

Spring 2023

- Developed a versatile model to simulate multi-wavelength transits of a Titan-like exoplanet, enabling the analysis of methane-like behavior in light curves to assess habitability potential
- Programmed light curve variations for different orbital radii and conducted research on atmospheric effects of various stellar spectral types on the modeled exoplanet

SKILLS

- Python
 - Linux
- CASA
 - LaTeX
- CARTA
 - ds9
- Mathematica
 - Conversational & written French
- SLURM

AWARDS & HONORS

Magna cum laude, CU Boulder

CU Boulder Chancellor’s Achievement Scholarship

Dean’s List: Fall 2019, Spring 2020, Spring 2022, Fall 2022, Spring 2023, Fall 2023, Spring 2024

May 2024

2019-2023

RELEVANT COURSEWORK

Formation & Dynamics of Planetary Systems | ASTR 3710

Observations & Instrumentation I & II | ASTR 3510 & 3520

Astrophysics I: Stellar & Interstellar | ASTR 3750

Planets, Moons, & Rings | ASTR 3750

Introductory Radio Astronomy
ASTR 5340 *audited at UVA*

Research Methods in Astronomy | ASTR 3400

Planets & Their Atmospheres | ASTR 3720

Astronomical Data Analysis | ASTR 3800

Becoming a Learning Assistant | EDUC 4610

Introduction to Astrochemistry
ASTR 5260 *audited at UVA*

OUTREACH & LEADERSHIP

Dark Skies, Bright Kids

Volunteer-run science outreach group serving elementary schools across Virginia

Fall 2024-Present

- Helped create accessible astronomy lesson plans for blind and low-vision students & taught science classes for grades 3–12 at the Virginia School for the Deaf and the Blind
- Presented planetarium shows to elementary school students through an after-school library program

President, CU Boulder Rotaract Club**2023-2024***Community service club in conjunction with Rotary Club*

- Coordinated monthly volunteer events to distribute hot meals and clothing to Boulder's homeless population
- Organized a fundraiser and collaborated with a university in Istanbul to support the construction of a field hospital in Turkey following the devastating earthquakes

Secretary, CU Boulder Rotaract Club**2022-2023**

REFERENCES**John Tobin, PhD**

Scientist at the National Radio Astronomy Observatory

jtobin@nrao.edu

Meredith MacGregor, PhD

Assistant Professor- Johns Hopkins University, Dept. of Physics & Astronomy

mmacgregor@jhu.edu

Noah Finkelstein, PhD

Professor- University of Colorado Boulder, Dept. of Physics

noah.finkelstein@colorado.edu